

LESSON 6-4 • SESSION 1 • PRACTICE • FORM A

Writing Numerical Expressions

Learning target – I can write a numerical expression that matches a situation. • 6.EE.A.2

REMEMBER

- A **numerical expression** has numbers and operations but **no equal sign**.
- “Each” or “per” means **equal groups**, so multiply. Totals of groups are **added**. “5 less than 12” is $12 - 5$.
- Use **parentheses** for a part done first: the sum of 4 and 6, tripled, is $3 \times (4 + 6)$.

WORKED EXAMPLE

Ms. Ortiz buys 4 pizzas at \$9 each and one \$6 salad. Write an expression for the total. Evaluate it.

Step 1 Pizzas are equal groups: 4×9 .

Step 2 Add the salad: $4 \times 9 + 6$.

Step 3 Evaluate: $36 + 6 = 42$.

Answer $4 \times 9 + 6 = \$42$

WORDS numerical expression (*expresión numérica*) • evaluate (*evaluar*) • product (*producto*)

PRACTICE

show your work • label units

- Write a numerical expression for each phrase. Do not evaluate.
 - 4 more than the product of 6 and 3 _____
 - 20 less than 75 _____
 - the sum of 5 and 9, tripled _____
 - the quotient of 48 and 8, plus 7 _____
- School Store Order.** A school store orders 5 boxes of pens at \$12 a box and 4 packs of paper at \$3 a pack. It also stacks notebooks in 10 rows of 10.
 - Write one expression for the total order cost. Evaluate it. _____ dollars
 - Write the notebooks as a power. Evaluate it. _____ notebooks

Show your work.

- Field Trip.** A field trip costs \$13 for each of 20 students, plus one \$25 bus fee. Write one expression for the total cost. Evaluate it.

Hint One product for the students, then add the fee.
- Square Garden.** A square garden has 15 plants along each side, filling the whole square. Write the number of plants as a power. Evaluate it.

Hint A square uses the same number twice.

Show your work.

Show your work.

Answer _____ dollars

Answer _____ plants

- Cracker Bags.** Leo fills 3 bags with 16 crackers each, then eats 5 crackers. He wrote $3 + 16 - 5$ for the crackers left. Justify or correct his expression.

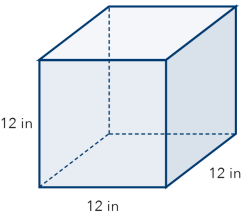
Hint Which operation matches the word “each”?

Answer _____ crackers

- 6. Cube Bin.** A cube-shaped bin is 12 inches on every edge. Write the number of 1-inch cubes that fill it as a power. Evaluate it.

Hint A cube has 3 equal dimensions.

Show your work.



Answer _____ cubes

- 7. Supply Change.** A class buys 12 notebooks at \$3 each and 20 pens at \$2 each. It pays with \$100.
- Write an expression for the total cost. _____
 - Write an expression for the change. Evaluate it. _____ dollars
 - Why do parentheses help in part b? _____

- 8. Tickets and Programs.** A group buys 4 tickets at \$9 each and 3 programs at \$6 each. Write one expression for the total cost. Evaluate it.

Hint Write a product for each item, then add.

Show your work.

Answer _____ dollars

FIND THE ERROR

explain the mistake, then fix it

- 9. Ines's Work.** Snacks cost \$3 each. Ines buys 4 snacks and one \$10 poster. She wrote $3 + 4 \times 10$ for the total. Explain her mistake. Then write and evaluate a correct expression.

TEST PRACTICE

choose the best answer

- 10.** "3 more than the product of 5 and 4" is written –
- A** $5 \times 4 + 3$ **B** $3 \times 5 + 4$ **C** $(3 + 5) \times 4$ **D** $5 + 4 \times 3$
- 11.** "15 less than 40" is written –
- A** $15 - 40$ **B** $40 - 15$ **C** $40 + 15$ **D** 15×40

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

24 C	43 D	54 T	285 R	30 X	14 Q	36 Z	1,728 U	72 P	225 O
#2a		#3	#4	#5	#6	#7b	#8		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____